SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:30**

Annexure A

Scenario-Based

Code of Conduct:

*I **D Varshini - 1912320007** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. Use Case Diagram Scenario

A **Smart Pharmacy Management System** enables customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage medicine inventory and user accounts. The interactions among system users are not clearly defined.

Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.

ANSWER :

Step 1: Identify Actors

1. Customer
2. Pharmacist
3. Administrator
4. Payment Gateway (External System)

Step 2: Identify Use Cases

Customer

- Register/Login
- Search Medicines
- Upload Prescription
- Place Medicine Order
- Make Online Payment
- Track Order

Pharmacist

- Verify Prescription
- Dispense Medicines
- Update Order Status

Administrator

- Manage Medicine Inventory
- Manage User Accounts
- Generate Reports

Payment Gateway

- Process Payment

Step 3: Relationships

<<include>> Relationships

- Place Medicine Order → Search Medicines
- Place Medicine Order → Make Online Payment
- Make Online Payment → Process Payment
- Dispense Medicines → Verify Prescription

<<extend>> Relationships

- Upload Prescription <<extend>> Place Medicine Order
    - (Only required for prescription medicines.)

Generalization

- Administrator and Pharmacist are specialized users of System User.

Step 4 : Use case diagram

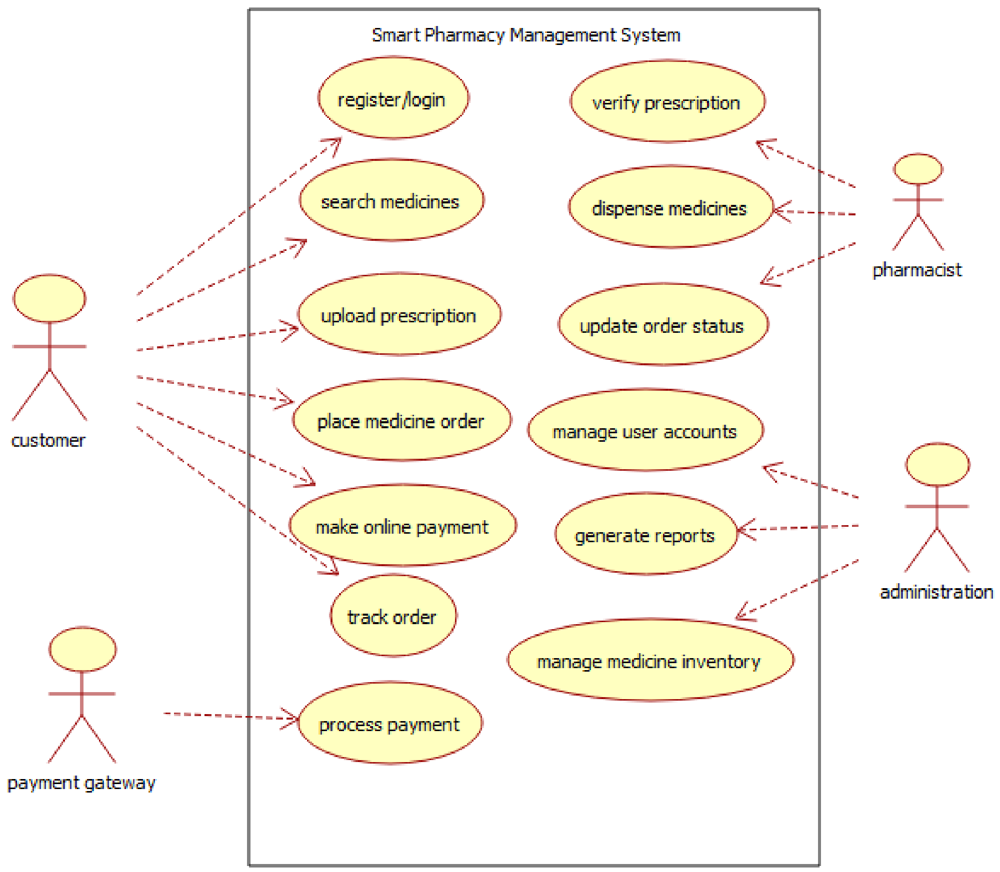

Fig 1: Use case diagram - Smart Pharmacy Management System

Explanation

- **Customer** searches medicines, uploads prescriptions (if needed), places orders, pays online, and tracks orders.
- **Pharmacist** verifies uploaded prescriptions before dispensing medicines and updates order status.
- **Administrator** manages inventory, user accounts, and system reports.
- **Payment Gateway** processes online payments.
- **<<include>>** is used for mandatory actions that are always part of another use case.
- **<<extend>>** is used for optional behavior (uploading a prescription only for prescription-required medicines).
- **Generalization** groups common functionalities under the parent actor **System User**.

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2. Class Diagram Scenario

A **Smart Hotel Management System** consists of entities such as Guest, Room, Reservation, Payment, and Receptionist. The existing design does not clearly define class attributes, operations, or relationships among these entities.

Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.

ANSWER:

1. Classes with Attributes and Methods

Class	**Attributes**	**Methods**
Guest	guestID, name, phone, email, address	register(), bookRoom(), cancelReservation()
Room	roomNo, roomType, price, status	checkAvailability(), updateStatus()
Reservation	reservationID, checkInDate, checkOutDate, bookingStatus	createReservation(), modifyReservation(), cancelReservation()
Payment	paymentID, amount, paymentMode, paymentStatus	makePayment(), generateReceipt()

Receptionist	employeeID, name, shift	assignRoom(), checkInGuest(), checkOutGuest()

2. Relationships

Relationship Type	Classes Involved	Multiplicity	Description
Association	Guest ↔ Reservation	1 Guest : 0..* Reservations	A guest can make multiple reservations.
Association	Receptionist ↔ Reservation	1 Receptionist : 0..* Reservations	Receptionist manages reservations.
Composition	Reservation ◆── Payment	1 Reservation : 1 Payment	Payment cannot exist without a reservation.
Inheritance	Person → Guest, Receptionist	One parent, two child classes	Guest and Receptionist inherit common details from Person.

3. Parent Class (Inheritance)

Person

- Attributes: personID, name, phone
- Methods: login(), logout()

Inherited Classes

- Guest
- Receptionist

4. Multiplicity Summary

From	To	Multiplicity
Guest	Reservation	1 → 0..*
Reservation	Room	1 → 1

Reservation	Payment	1 → 1
Receptionist	Reservation	1 → 0..*

Relationship Symbols

- Association (—) → Guest ↔ Reservation, Receptionist ↔ Reservation
- Composition (◆) → Reservation ◆── Payment
- Inheritance (△) → Person △── Guest, Receptionist

Class Diagram :

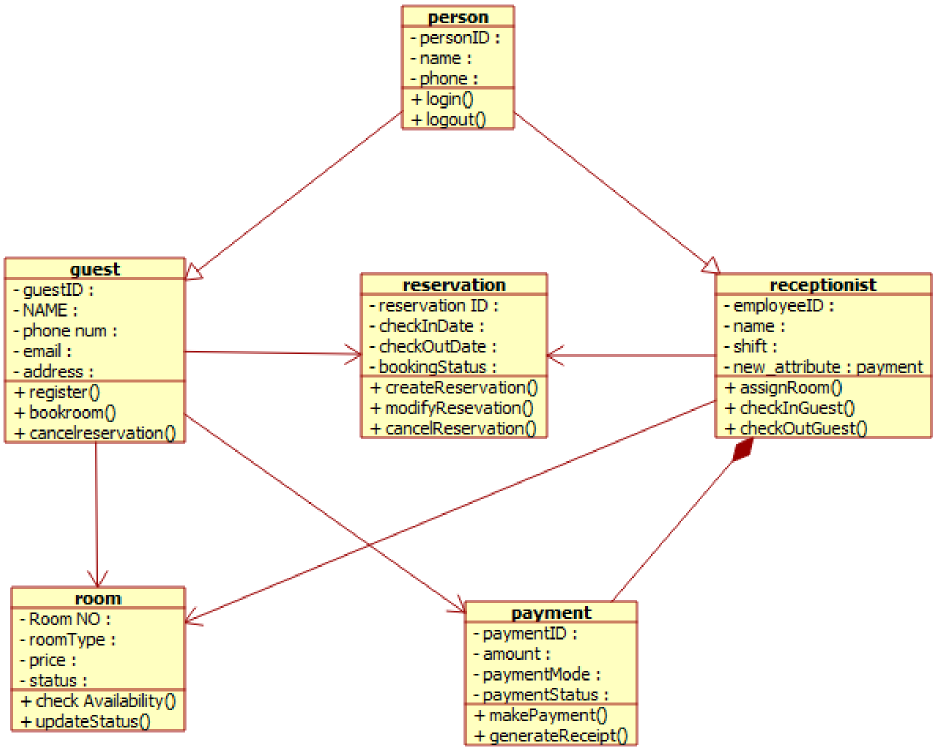

Fig 2 : Class Diagram : **Smart Hotel Management System**

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3. Sequence Diagram Scenario

In an **Online Banking System**, a customer logs in, transfers funds, verifies the transaction using OTP, and receives a transaction confirmation. The communication among the customer, authentication service, banking server, and database is incomplete.

Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.

ANSWER:

Step 1: Identify Participating Objects

1. Customer
2. Authentication Service
3. Banking Server
4. Database

Step 2: Sequence of Operations

1. Customer enters username and password.
2. Authentication Service validates the login credentials with the Database.
3. Database returns the validation result.
4. Authentication Service grants access to the Customer.
5. Customer initiates Fund Transfer.
6. Banking Server requests an OTP from the Authentication Service.
7. Authentication Service sends the OTP to the Customer.
8. Customer enters the OTP.
9. Authentication Service verifies the OTP.
10. Banking Server processes the fund transfer.
11. Banking Server updates the Database.
12. Database confirms successful transaction.
13. Banking Server sends a Transaction Confirmation to the Customer.

Explanation

- The Customer logs into the system.
- The Authentication Service verifies credentials using the Database.
- After successful login, the customer requests a fund transfer.
- The Banking Server requests OTP verification through the Authentication Service.
- The customer enters the OTP, which is verified.
- If the OTP is valid, the Banking Server updates the transaction in the Database.

- Finally, the customer receives a transaction confirmation.

Step 3: Sequence Diagram

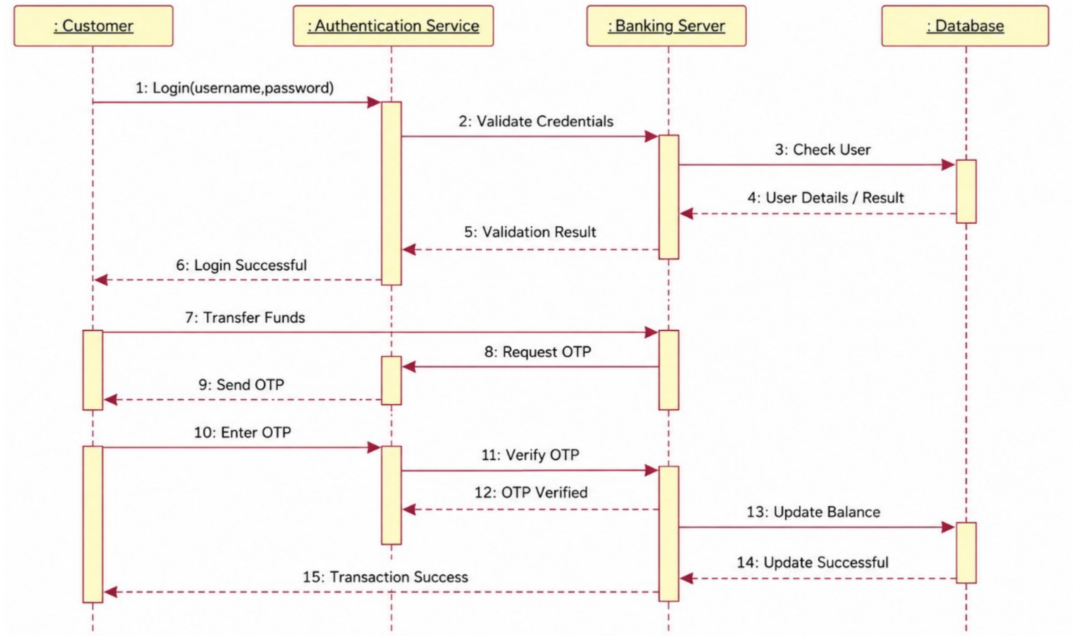

Fig 3 : Sequence diagram - **Online Banking System**

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4. Activity Diagram Scenario

A **Passport Application System** allows applicants to submit applications, upload supporting documents, complete identity verification, pay processing fees, and receive passports. The workflow does not clearly represent approval, rejection, or concurrent activities.

Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.

ANSWER :

Step 1: Identify Activities

- Submit Passport Application
- Upload Supporting Documents
- Verify Identity
- Pay Processing Fee
- Verify Documents

- Approve Application
- Reject Application
- Print Passport
- Dispatch Passport
- Receive Passport

Step 2: Identify Decision Point

Decision: Are the submitted documents and identity verification valid?

- Yes → Approve the application.
- No → Reject the application and notify the applicant.

Step 3: Identify Parallel Activities (Fork & Join)

Fork (Concurrent Activities):

- Identity Verification
- Processing Fee Payment

Join:

- Continue only after both identity verification and fee payment are completed.

Explanation

- The process begins when the applicant submits the passport application and uploads the required documents.
- The system performs identity verification and processing fee payment simultaneously using a Fork.
- A Join node ensures both activities are completed before moving forward.
- A Decision node checks whether the documents and identity verification are valid.
- If approved, the passport is printed, dispatched, and delivered to the applicant.
- If rejected, the applicant is notified, and the process ends.
- The workflow concludes at the End Node.

Step 4: Activity Diagram

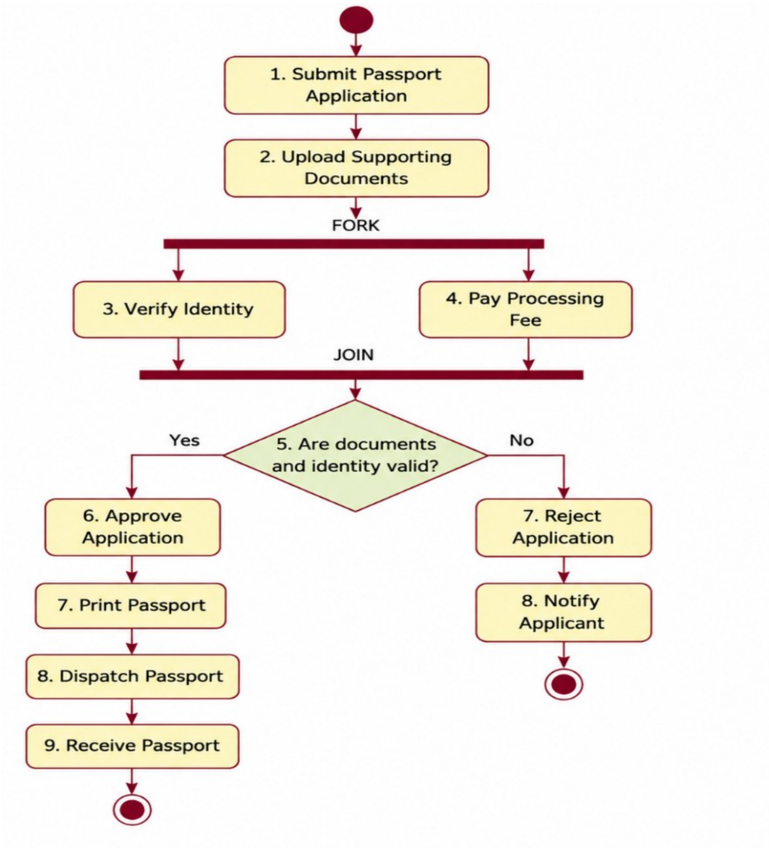

Fig 4 : Activity diagram - **Passport Application System**

Step 5: Explanation

- The process begins when the applicant submits the passport application and uploads the required documents.
- The system performs identity verification and processing fee payment simultaneously using a Fork.
- A Join node ensures both activities are completed before moving forward.
- A Decision node checks whether the documents and identity verification are valid.
- If approved, the passport is printed, dispatched, and delivered to the applicant.
- If rejected, the applicant is notified, and the process ends.
- The workflow concludes at the End Node.

5. State Diagram Scenario

A **Smart Elevator Control System** operates through states such as Idle, Door Open, Door Closed, Moving Up, Moving Down, Emergency Stop, and Maintenance Mode. The transitions between these states are not clearly specified.

Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the elevator's behavior.

ANSWER :

Step 1: Identify States

- Idle
- Door Open
- Door Closed
- Moving Up
- Moving Down
- Emergency Stop
- Maintenance Mode

Step 2: Identify Triggering Events

- Floor button pressed
- Door open button pressed
- Door close button pressed
- Destination floor selected
- Elevator reaches destination
- Emergency button pressed
- Emergency cleared
- Maintenance mode enabled
- Maintenance completed

Step 3: Identify State Transitions

- **Idle → Door Open** : Passenger calls elevator / Door Open button pressed.
- **Door Open → Door Closed** : Door Close button pressed or timeout occurs.
- **Door Closed → Moving Up** : Destination floor is above the current floor.
- **Door Closed → Moving Down** : Destination floor is below the current floor.

- **Moving Up → Door Open** : Elevator reaches the selected upper floor.
- **Moving Down → Door Open** : Elevator reaches the selected lower floor.
- **Door Open → Idle** : No pending floor requests.
- **Moving Up → Emergency Stop** : Emergency button pressed.
- **Moving Down → Emergency Stop** : Emergency button pressed.
- **Emergency Stop → Idle** : Emergency cleared and system reset.
- **Any State → Maintenance Mode** : Maintenance mode enabled.
- **Maintenance Mode → Idle** : Maintenance completed.

Step 4: State Transition Diagram

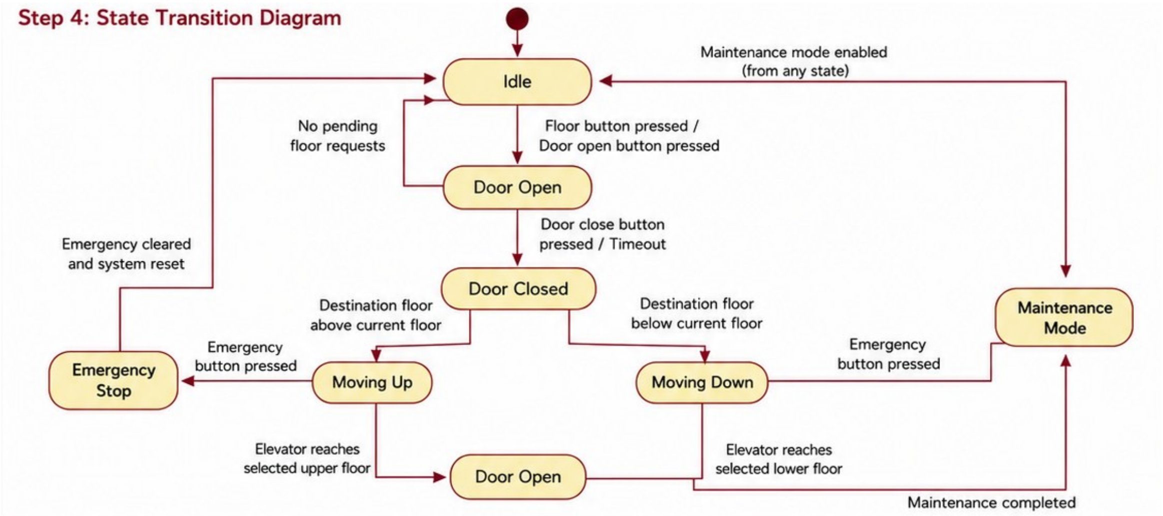

Fig 5 : State Transition diagram - **Smart Elevator Control System**

Step 5: Explanation

- The elevator starts in the Idle state.
- When a passenger requests the elevator, the Door Open state is entered.
- After the doors close, the elevator moves Up or Down depending on the selected floor.
- When the destination floor is reached, the doors open again.
- If there are no more requests, the elevator returns to the Idle state.
- Pressing the emergency button at any time while moving changes the state to Emergency Stop.
- Enabling maintenance mode from any state switches the system to Maintenance Mode.
- After maintenance or emergency recovery, the elevator returns to the Idle state.

6. Deployment Diagram Scenario

A **Smart Home Automation System** uses smart sensors, IoT controllers, a home gateway, cloud servers, and a mobile application to monitor and control lighting, security, and appliances remotely. The deployment architecture does not clearly illustrate hardware nodes, deployed software, or communication channels.

Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.

ANSWER :

Step 1: Identify Hardware Nodes

- Smart Sensors
- IoT Controller
- Home Gateway (Wi-Fi Router/Hub)
- Cloud Server
- Mobile Device (Smartphone)

Step 2: Identify Software Artifacts

- Sensor Firmware
- IoT Control Software
- Home Automation Application
- Cloud Management Service
- Mobile Application

Step 3: Identify Communication Links

- Smart Sensors ↔ IoT Controller : Sensor data transmission.
- IoT Controller ↔ Home Gateway : Sends control commands and receives sensor data.
- Home Gateway ↔ Cloud Server : Internet communication using Wi-Fi/Ethernet.
- Cloud Server ↔ Mobile Application : Remote monitoring and control.
- Mobile Application ↔ Cloud Server ↔ Home Gateway : User sends commands to control home devices.

Step 4: Deployment Diagram

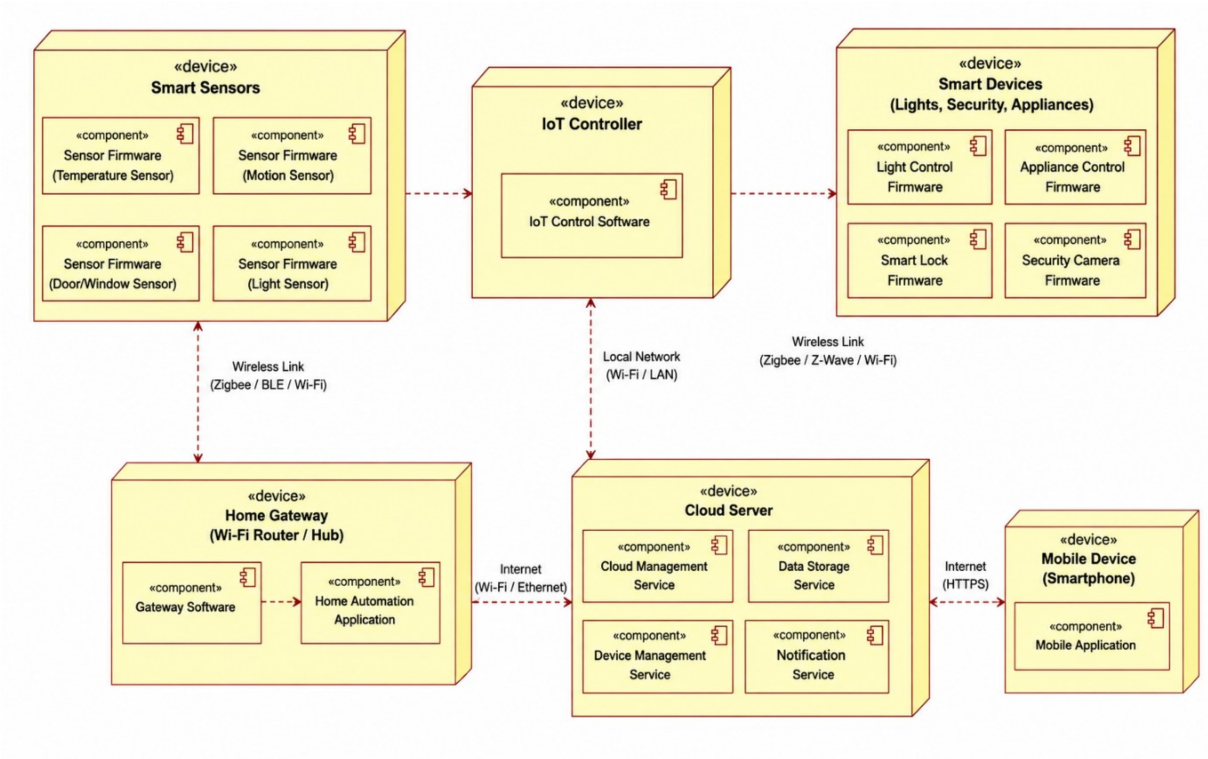

Fig 6 : Deployment diagram - **Smart Home Automation System**

Step 5: Explanation

- Smart Sensors collect environmental data such as temperature, motion, and light intensity.
- The IoT Controller processes sensor data and controls smart devices.
- The Home Gateway connects the local smart home network to the internet.
- The Cloud Server stores data, processes requests, and enables remote access.
- The Mobile Application allows users to monitor and control lighting, security systems, and appliances from anywhere.
- Communication occurs through wireless links, local Wi-Fi/LAN, and the Internet, ensuring seamless interaction between all components.

RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand